SOLVED PRACTICE WORKBOOK

Geography 01 - The Earth and the Universe / India Location and Extent - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Geography 01 - The Earth and the Universe / India Location and Extent

BASIC MCQS / REMEDIATION

29. Diagnostic design and rotation rule

The 40 diagnostic questions are balanced across the two halves: Q1-Q20 test planetary geography and Q21-Q40 test India Location and Extent. Q41-Q48 are remedials. Correct options rotate strictly **A -> B -> C -> D** without interruption.

Q1. Which sequence correctly places Earth in its largest-to-smallest astronomical hierarchy?

- A. Universe -> Milky Way galaxy -> Solar System -> Earth
- B. Milky Way -> universe -> Earth -> Solar System
- C. Solar System -> universe -> Milky Way -> Earth
- D. Universe -> Solar System -> Milky Way -> Earth

Answer: A. Universe -> Milky Way galaxy -> Solar System -> Earth

Explanation: The universe contains galaxies; the Milky Way contains the Solar System; Earth is one planet within it.

Q2. Which statement is the safest description of the Big Bang?

- A. It was a bomb-like explosion from one point into empty space.
- B. It created only the Milky Way while the rest of space already existed.
- C. It is proved by Earth's seasonal cycle.
- D. It describes expansion and cooling from an extremely hot, dense early state.

Answer: D. It describes expansion and cooling from an extremely hot, dense early state.

Explanation: NASA's current cosmology overview places the early universe at about 13.8 billion years and treats expansion, nucleosynthesis and the cosmic microwave background as linked evidence.

Q3. Why do high-mass stars normally have shorter lifespans than low-mass stars?

- A. They contain no hydrogen.
- B. They rotate only clockwise.
- C. Their cores sustain much faster nuclear-fusion rates.
- D. They are always nearer Earth.

Answer: C. Their cores sustain much faster nuclear-fusion rates.

Explanation: Greater mass produces higher core temperature and pressure, so fuel is consumed disproportionately faster despite the larger fuel stock.

Q4. Which pairing is correct?

- A. Spiral galaxy - disc with arms and a central bulge
- B. Milky Way - an isolated planet
- C. Irregular galaxy - perfectly symmetrical disc
- D. Elliptical galaxy - always rich in spiral arms

Answer: A. Spiral galaxy - disc with arms and a central bulge

Explanation: Galaxy classification is morphological. The Milky Way is a barred spiral; irregular galaxies lack a regular spiral or elliptical form.

Q5. Which sequence best represents nebular formation of the Solar System?

- A. Planet formation -> nebula -> galaxy formation -> collapse
- B. Cloud collapse -> rotating disc -> accretion -> differentiated planets
- C. Accretion -> Big Bang -> cloud collapse -> Sun

- D. Earth cooling -> Sun formation -> cosmic microwave background

Answer: B. Cloud collapse -> rotating disc -> accretion -> differentiated planets

Explanation: Gravity collapsed gas and dust; angular momentum flattened the material into a disc; accretion assembled planetesimals and planets.

Q6. Which group contains only terrestrial planets?

- A. Jupiter, Saturn, Uranus and Neptune
- B. Earth, Jupiter, Saturn and Mars
- C. Mercury, Venus, Uranus and Neptune
- D. Mercury, Venus, Earth and Mars

Answer: D. Mercury, Venus, Earth and Mars

Explanation: The four inner planets are relatively small and rocky/metallic; the outer four are giant planets.

Q7. Which distinction is correct?

- A. A meteor is a dwarf planet.
- B. A meteoroid is in space; a meteor is the luminous atmospheric phenomenon; a meteorite reaches the ground.
- C. A meteorite is always a comet.
- D. A meteoroid exists only after reaching the ground.

Answer: B. A meteoroid is in space; a meteor is the luminous atmospheric phenomenon; a meteorite reaches the ground.

Explanation: The terms identify location/stage, not three unrelated types of planet.

Q8. Which rotation statement is correct?

- A. All planets rotate in precisely the same manner.
- B. Uranus has no axial tilt.
- C. Earth rotates east to west.
- D. Venus rotates retrograde relative to most planets.

Answer: D. Venus rotates retrograde relative to most planets.

Explanation: NASA describes Venus as rotating backward relative to most planets; Uranus has an extreme axial tilt and Earth rotates west to east.

Q9. Why is Earth called an oblate spheroid?

- A. Rotation produces equatorial bulging and polar flattening.
- B. Ocean trenches pull the Equator inward.
- C. Seasons alternately flatten the hemispheres.
- D. The Moon makes Earth a perfect sphere.

Answer: A. Rotation produces equatorial bulging and polar flattening.

Explanation: The geometric flattening is small but measurable and is tied to rotation.

Q10. What is the geoid?

- A. The legal boundary of the oceans
- B. The line joining all mountain summits
- C. A gravity-defined equipotential surface approximating global mean sea level
- D. A perfectly smooth sphere with no gravity variation

Answer: C. A gravity-defined equipotential surface approximating global mean sea level

Explanation: The geoid is a physical gravity reference; an ellipsoid is a convenient mathematical approximation.

Q11. Which relation is correct?

- A. Rotation alone causes seasons.

- B. Earth-Sun distance alone causes seasons.
- C. Revolution alone causes day and night.
- D. Rotation causes day and night; revolution plus tilt causes seasons.

Answer: D. Rotation causes day and night; revolution plus tilt causes seasons.

Explanation: Tilt changes solar angle and day length as Earth revolves; daily rotation produces the day-night cycle.

Q12. Why is leap-year adjustment required?

- A. Earth rotates 366 times every year.
- B. A tropical year is about 365.24 days, not exactly 365 days.
- C. The Moon adds one full day annually.
- D. The International Date Line shifts by one day every four years.

Answer: B. A tropical year is about 365.24 days, not exactly 365 days.

Explanation: Calendar rules approximate the fractional day accumulated each year; century exceptions refine the simple four-year rule.

Q13. Which statement about seasons is correct?

- A. Aphelion always causes winter in both hemispheres.
- B. The hemispheres experience the same season simultaneously.
- C. The Sun is overhead at the poles on every solstice.
- D. Perihelion occurs during Northern Hemisphere winter, showing that distance is not the primary cause of seasons.

Answer: D. Perihelion occurs during Northern Hemisphere winter, showing that distance is not the primary cause of seasons.

Explanation: Opposite seasons in the two hemispheres and January perihelion demonstrate the controlling role of tilt, not distance.

Q14. On the June solstice, where is the Sun overhead at noon?

- A. The Arctic Circle
- B. The Tropic of Cancer
- C. The Tropic of Capricorn
- D. The Equator

Answer: B. The Tropic of Cancer

Explanation: The subsolar point reaches about 23.5°N at the June solstice.

Q15. Which statement about longitude is correct?

- A. All parallels are great circles.
- B. A longitude degree has the same linear length everywhere.
- C. Longitude measures angular distance north or south.
- D. Meridians converge toward the poles.

Answer: D. Meridians converge toward the poles.

Explanation: Meridian convergence makes a degree of longitude shorter away from the Equator.

Q16. A place 30° east of another place is ahead in local solar time by:

- A. Thirty minutes
- B. Two hours
- C. Four hours
- D. One day

Answer: B. Two hours

Explanation: At 15° per hour, a 30° difference equals two hours.

Q17. What is the correct International Date Line travel rule?

- A. Crossing westward subtracts one day.
- B. No date change occurs near 180°.
- C. Crossing westward adds one day; crossing eastward subtracts one day.
- D. Both directions add one day.

Answer: C. Crossing westward adds one day; crossing eastward subtracts one day.

Explanation: The rule reconciles calendar dates after accumulating longitudinal time differences; the line deviates from exact 180°.

Q18. Which sequence best explains an aurora?

- A. Ocean current -> evaporation -> cloud -> aurora
- B. Moonlight -> ice reflection -> polar cloud -> lightning
- C. Earthquake wave -> ionosphere -> sunlight
- D. Solar wind/CME -> magnetospheric interaction -> particle precipitation -> gas excitation -> light

Answer: D. Solar wind/CME -> magnetospheric interaction -> particle precipitation -> gas excitation -> light

Explanation: Aurora is an upper-atmospheric solar-terrestrial coupling phenomenon, not weather or reflected light.

Q19. Which gas-colour pairing is broadly correct?

- A. Nitrogen - only yellow at all heights
- B. Atomic oxygen - common green and high-altitude red
- C. Oxygen - black
- D. Carbon dioxide - the only auroral colour source

Answer: B. Atomic oxygen - common green and high-altitude red

Explanation: NASA identifies oxygen as the source of green and red emissions; nitrogen contributes blue, purple and pink.

Q20. What does the S-wave shadow reveal?

- A. A liquid outer core that shear waves cannot traverse
- B. A hollow mantle
- C. A gaseous inner core
- D. A liquid crust

Answer: A. A liquid outer core that shear waves cannot traverse

Explanation: S-waves travel through solids but not liquids; P-wave refraction supplies complementary evidence.

Q21. Which is an example of India's situation rather than its site?

- A. The physical ground occupied by the peninsula
- B. The rock type beneath a city
- C. Its relationship to West Asian and South-East Asian sea routes
- D. The local slope of a settlement

Answer: C. Its relationship to West Asian and South-East Asian sea routes

Explanation: Situation is relational location; site is the physical ground occupied.

Q22. Which coordinate envelope is the standard mainland extent of India?

- A. 8°4'S-37°6'S and 68°7'E-97°25'E
- B. 8°4'N-37°6'N and 68°7'E-97°25'E
- C. 6°45'N-37°6'N and 68°7'W-97°25'W
- D. 23°30'N-82°30'N and 68°7'E-97°25'E

Answer: B. 8°4'N-37°6'N and 68°7'E-97°25'E

Explanation: The familiar four coordinates refer to mainland India; island territory extends farther south.

Q23. Which distinction is correct?

- A. Indira Point is mainland south.
- B. Kanyakumari is the southernmost point of the entire Union.
- C. Kanyakumari is mainland south; Indira Point is territorial south.
- D. Both points are in Lakshadweep.

Answer: C. Kanyakumari is mainland south; Indira Point is territorial south.

Explanation: The stem must specify mainland or entire territory; confusing the categories is a standard trap.

Q24. Which is the correct west-to-east Tropic of Cancer order?

- A. Rajasthan -> Gujarat -> Madhya Pradesh -> Odisha -> Jharkhand -> Assam -> Tripura -> Mizoram
- B. Gujarat -> Maharashtra -> Madhya Pradesh -> Chhattisgarh -> Odisha -> West Bengal -> Assam -> Manipur
- C. Gujarat -> Rajasthan -> Madhya Pradesh -> Chhattisgarh -> Jharkhand -> West Bengal -> Tripura -> Mizoram
- D. Gujarat -> Rajasthan -> Uttar Pradesh -> Bihar -> Jharkhand -> West Bengal -> Tripura -> Mizoram

Answer: C. Gujarat -> Rajasthan -> Madhya Pradesh -> Chhattisgarh -> Jharkhand -> West Bengal -> Tripura -> Mizoram

Explanation: The line crosses eight states; Odisha, Maharashtra, Uttar Pradesh, Bihar and Assam are common distractors.

Q25. Which statement about the Tropic of Cancer is safest?

- A. It is a solar-geometric reference, not a climatic wall.
- B. It divides India into two exactly equal areas.
- C. Everything south is uniformly equatorial.
- D. It determines monsoon rainfall without relief or circulation.

Answer: A. It is a solar-geometric reference, not a climatic wall.

Explanation: Latitude matters, but altitude, relief, circulation, continentality and ocean influence prevent a sharp climatic boundary.

Q26. Why is Indian Standard Time UTC+5:30?

- A. India lies 5.5° east of Greenwich.
- B. The International Date Line passes through Mirzapur.
- C. 82.5° × four minutes per degree = 330 minutes.
- D. The Tropic of Cancer is 5.5 hours ahead.

Answer: C. 82.5° × four minutes per degree = 330 minutes.

Explanation: The standard meridian at 82°30'E converts to five hours thirty minutes ahead of Greenwich/UTC.

Q27. Which statement about 82°30'E is correct?

- A. It is India's westernmost longitude.
- B. It is the Tropic of Cancer.
- C. It is a political state boundary.
- D. It is near the mainland's central longitude, not its exact midpoint.

Answer: D. It is near the mainland's central longitude, not its exact midpoint.

Explanation: The arithmetic midpoint of 68°7'E and 97°25'E is about 82°46'E.

Q28. India's east-west local solar-time spread is approximately:

- A. 117 minutes, conventionally about two hours
- B. 29 minutes

- C. Twenty-four hours
- D. 330 minutes

Answer: A. 117 minutes, conventionally about two hours

Explanation: The 29°18' span multiplied by four minutes per degree gives about 117 minutes.

Q29. Why are India's similar angular north-south and east-west spans unequal in kilometres?

- A. Latitude degrees become zero in India.
- B. Earth stops rotating over the peninsula.
- C. Longitude degrees contract as meridians converge poleward.
- D. The Tropic of Cancer bends around mountains.

Answer: C. Longitude degrees contract as meridians converge poleward.

Explanation: A degree of latitude remains nearly constant; longitude degree length varies with latitude.

Q30. What does the 2025 coastline revision to 11,098.81 km primarily show?

- A. India acquired new territory.
- B. Finer GIS/high-water-line methodology captured more coastal and island detail.
- C. India's land area automatically increased.
- D. All Coastal Regulation Zone boundaries shifted automatically.

Answer: B. Finer GIS/high-water-line methodology captured more coastal and island detail.

Explanation: The revised measurement is scale- and method-dependent; it is not territorial growth.

Q31. Which statement about India's land neighbours is correct?

- A. The official list includes Afghanistan under India's claim, although there is no functioning direct crossing.
- B. India has only five official land neighbours.
- C. Maldives shares a Himalayan frontier.
- D. Sri Lanka is a land neighbour.

Answer: A. The official list includes Afghanistan under India's claim, although there is no functioning direct crossing.

Explanation: The official MHA list has seven entries; operational accessibility and legal claim are distinct.

Q32. Which maritime pairing is correct?

- A. Bhutan - Gulf of Mannar
- B. Sri Lanka - Palk Strait and Gulf of Mannar
- C. Sri Lanka - Arabian Desert
- D. Maldives - Ten Degree Channel

Answer: B. Sri Lanka - Palk Strait and Gulf of Mannar

Explanation: Palk Strait and Gulf of Mannar separate India and Sri Lanka; Maldives lies south-west of Lakshadweep.

Q33. Why is India considered a subcontinent?

- A. Continental scale and relative physical coherence coexist with deep internal diversity.
- B. It is completely isolated from Asia.
- C. It is culturally homogeneous.
- D. It is a separate tectonic plate today in every sense.

Answer: A. Continental scale and relative physical coherence coexist with deep internal diversity.

Explanation: Scale, a recognisable Himalayan-oceanic frame, physiographic linkage and diversity support the term; permeability qualifies isolation.

Q34. Which claim about Indian Ocean centrality is defensible?

- A. India legally owns all three chokepoints.

- B. Map centrality guarantees trade dominance.
- C. The Bay of Bengal lies west of India.
- D. India is strategically affected by routes through Hormuz, Bab-el-Mandeb and Malacca but does not automatically control them.

Answer: D. India is strategically affected by routes through Hormuz, Bab-el-Mandeb and Malacca but does not automatically control them.

Explanation: Centrality is relational and must be converted into capability by ports, institutions, partnerships and logistics.

Q35. Which island statement is correct?

- A. All Indian islands are coral atolls.
- B. Lakshadweep is mainly coral, while Andaman-Nicobar has a different tectonic-geological setting.
- C. The Ten Degree Channel separates India and Sri Lanka.
- D. Andaman Sea lies west of Lakshadweep.

Answer: B. Lakshadweep is mainly coral, while Andaman-Nicobar has a different tectonic-geological setting.

Explanation: The two island systems differ in basin, geology, hazards and ecology; Ten Degree Channel separates Andaman and Nicobar groups.

Q36. Which statement best describes the Himalaya and surrounding seas?

- A. They are selective filters and connectors, not absolute barriers.
- B. Mountains determine culture without human agency.
- C. Seas only isolate and never connect.
- D. They prevented all historical exchange.

Answer: A. They are selective filters and connectors, not absolute barriers.

Explanation: Passes, valleys and maritime routes channel interaction even as high relief and distance raise movement costs.

Q37. Which official-value pairing is correct?

- A. Area 11,098.81 sq km; land border 3,287,263 km
- B. Area 15,106.7 sq km; land border 7,516.6 km
- C. Area 3,287,263 sq km; international land border 15,106.7 km
- D. Area 2.4 sq km; land border 82.5 km

Answer: C. Area 3,287,263 sq km; international land border 15,106.7 km

Explanation: Area, land-border length and coastline length are different dimensions and must not be interchanged.

Q38. The Ten Degree Channel separates:

- A. Lakshadweep from Maldives
- B. The Andaman group from the Nicobar group
- C. India from Sri Lanka
- D. The Arabian Sea from the Bay of Bengal

Answer: B. The Andaman group from the Nicobar group

Explanation: The channel is an internal Andaman-Nicobar map reference; Palk Strait and Gulf of Mannar relate to Sri Lanka.

Q39. Which statement best connects India's location to climate and culture?

- A. Oceanic location makes every coast climatically identical.
- B. The Himalaya prevented all cultural contact.
- C. Latitude alone determines every crop and culture.

- D. Latitude and oceanic situation set broad conditions, while relief, circulation, routes and human agency shape regional outcomes.

Answer: D. Latitude and oceanic situation set broad conditions, while relief, circulation, routes and human agency shape regional outcomes.

Explanation: A strong answer recognises broad geographic conditions without converting them into deterministic outcomes.

Q40. Which is the safest UPSC map method for India-location answers?

- A. Copy an unverified internet boundary.
- B. Draw exact legal borders from memory.
- C. Omit island territories in every answer.
- D. Use a schematic not-to-scale outline, label only relevant features and never improvise disputed boundaries.

Answer: D. Use a schematic not-to-scale outline, label only relevant features and never improvise disputed boundaries.

Explanation: The purpose of a GS-I map is analytical orientation, not unsourced legal cartography.

30. Remedial MCQs

Q41. A learner says: 'Big Bang means matter exploded from a point into empty space.' What is the best correction?

- A. It describes expansion of space from a hot, dense early state; the point-explosion analogy is misleading.
- B. It describes Earth's volcanic origin.
- C. It is another name for a supernova.
- D. It occurred when the Solar System formed.

Answer: A. It describes expansion of space from a hot, dense early state; the point-explosion analogy is misleading.

Explanation: The correction separates cosmological expansion from an ordinary explosion within pre-existing space.

Q42. A learner says: 'January winter proves Earth is farthest from the Sun.' What is the best correction?

- A. Earth is near perihelion in early January; axial tilt, not distance, controls the hemispheric seasons.
- B. Earth has no elliptical orbit.
- C. January is winter in both hemispheres.
- D. Day length is unrelated to axial tilt.

Answer: A. Earth is near perihelion in early January; axial tilt, not distance, controls the hemispheric seasons.

Explanation: Opposite hemispheric seasons and January perihelion defeat the distance theory.

Q43. When it is Monday in Anadyr, what day is it most defensibly in Nome across the date line?

- A. Tuesday
- B. Monday everywhere
- C. Sunday
- D. Wednesday

Answer: C. Sunday

Explanation: Anadyr lies west of the International Date Line and is a calendar day ahead of Nome; the 2025 PYQ's Monday-Tuesday claim was therefore false.

Q44. Which inference from S- and P-wave behaviour is correct?

- A. The mantle is an ocean.
- B. The whole core is liquid.
- C. The outer core is liquid and the inner core is solid.
- D. P-waves cannot cross liquids.

Answer: C. The outer core is liquid and the inner core is solid.

Explanation: S-wave absence identifies liquid material; P-wave refraction and inner-core transmission support a solid inner core.

Q45. A learner uses 6°45'N as India's mainland southern limit. What is the correction?

- A. 6°45'N is the Standard Meridian.
- B. 6°45'N refers to the island extension; the mainland begins near 8°4'N.
- C. The mainland begins at 23°30'N.
- D. India lies south of the Equator.

Answer: B. 6°45'N refers to the island extension; the mainland begins near 8°4'N.

Explanation: Mainland and full-territory categories must be stated before quoting coordinates.

Q46. Which calculation correctly derives the approximate east-west solar-time spread?

- A. 82°30' × 4 minutes ≈ 117 minutes
- B. 29°18' ÷ 24 ≈ 330 minutes
- C. 29°18' × 4 minutes per degree ≈ 117 minutes
- D. 97°25' − 8°4' ≈ two hours

Answer: C. 29°18' × 4 minutes per degree ≈ 117 minutes

Explanation: Use the longitudinal span, not the Standard Meridian or mixed latitude-longitude values.

Q47. A learner says: 'The revised coastline proves India gained land.' What is the correction?

- A. The national area doubled.
- B. Every high-tide line was legally redrawn.
- C. Land borders were added to the coastline.
- D. The increase records finer-scale measurement of coastal complexity and islands, not territorial expansion.

Answer: D. The increase records finer-scale measurement of coastal complexity and islands, not territorial expansion.

Explanation: The coastline paradox explains why measured length grows as finer detail is captured.

Q48. Which sentence is the safest strategic conclusion?

- A. India's location creates opportunities and constraints whose outcomes depend on capacity, institutions and cooperation.
- B. India controls every Indian Ocean route.
- C. The Himalaya makes India completely isolated.
- D. Geography fixes development outcomes regardless of policy.

Answer: A. India's location creates opportunities and constraints whose outcomes depend on capacity, institutions and cooperation.

Explanation: This formulation is relational and anti-determinist while retaining the real value of location.

PYQS AND ANSWER PRACTICE

31. PYQ audit ledger and answer-key status

Route	Local official paper	Key/solution status
Prelims 2019 Q20	`books/more_previous_papers/csp-p1.pdf`	Official key not held locally; answer below is explicitly inferred
GS-I 2021 Q8	Local official GS-I paper	UPSC supplies no model solution; independent model below
Prelims 2022 Q29	`books/more_previous_papers/GENERAL STUDIES PAPER I.pdf`	Official key not held locally; answer below is explicitly inferred
Prelims 2024 Q15, Q32	Local official Set-A paper	Local official Set-A answer key: D and D
GS-I 2024 Q15	Local official GS-I paper	UPSC supplies no model solution; independent model below
Prelims 2025 Q33, Q77	Local official Set-A paper	Local official Set-A answer key: B and A

The repository routes exactly these eight genuinely relevant demands. No extra question is labelled as a verified PYQ merely because it resembles the topic.

32. UPSC Prelims 2019 Q20 - June solstice

Exact local-paper wording: On 21st June, the Sun:

- A. does not set below the horizon at the Arctic Circle
- B. does not set below the horizon at Antarctic Circle
- C. shines vertically overhead at noon on the Equator
- D. shines vertically overhead at the Tropic of Capricorn

INFERRED ANSWER - NOT OFFICIALLY VERIFIED LOCALLY: A (high confidence).

Statement-level explanation: On the June solstice the Northern Hemisphere tilts toward the Sun; the Arctic Circle can receive 24-hour daylight. The overhead Sun is at the Tropic of Cancer, not the Equator or Tropic of Capricorn. The Antarctic Circle experiences polar night.

33. UPSC GS-I 2021 Q8 - India as a subcontinent

Exact local-paper wording: “Why is India considered as a subcontinent? Elaborate your answer.” (10 marks, 150 words)

Independent model solution

India is considered a subcontinent because it combines continental scale with a relatively coherent physical frame within Asia. Its 3.287 million sq km area, Himalayan arc, Indo-Gangetic-Brahmaputra plains, peninsular plateau, long coast and island extensions form a large, internally connected geographic unit. The Himalaya and surrounding seas provide relative enclosure, while river systems, monsoon linkages and the plains-plateau-peninsula structure create physiographic coherence. Yet this scale also contains major climatic, ecological, linguistic and cultural diversity. Passes, valleys and Arabian Sea-Bay of Bengal routes historically connected the region with Central Asia, West Asia, East Africa and South-East Asia. Therefore, “subcontinent” does not imply isolation or homogeneity; it denotes a large and distinctive subdivision of Asia whose relative unity coexists with permeability and internal diversity.

Why this earns marks: It answers “why”, uses named spatial evidence, explains coherence and qualifies both isolation and cultural homogeneity.

34. UPSC Prelims 2022 Q29 - timing of the longest day

Exact local-paper wording: In the northern hemisphere, the longest day of the year normally occurs in the:

- A. First half of the month of June
- B. Second half of the month of June
- C. First half of the month of July
- D. Second half of the month of July

INFERRED ANSWER - NOT OFFICIALLY VERIFIED LOCALLY: B (high confidence).

Explanation: The June solstice occurs around 20-21 June, in the second half of June, and gives the Northern Hemisphere its maximum day length.

35. UPSC Prelims 2024 Q15 - latitude and day length

Exact local-paper wording: On June 21 every year, which of the following latitude(s) experience(s) a sunlight of more than 12 hours?

1. Equator
2. Tropic of Cancer
3. Tropic of Capricorn
4. Arctic Circle

- A. 1 only
- B. 2 only
- C. 3 and 4
- D. 2 and 4

OFFICIAL LOCAL SET-A KEY: D.

Statement-level explanation: The Equator is approximately 12 hours; the Tropic of Cancer lies in the hemisphere tilted toward the Sun and exceeds 12 hours; the Tropic of Capricorn is below 12 hours; the Arctic Circle can receive 24-hour daylight.

36. UPSC Prelims 2024 Q32 - giant and dwarf stars

Exact local-paper wording: Consider the following statements:

Statement-I: Giant stars live much longer than dwarf stars. Statement-II: Compared to dwarf stars, giant stars have a greater rate of nuclear reactions.

- A. Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
- B. Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I
- C. Statement-I is correct, but Statement-II is incorrect
- D. Statement-I is incorrect, but Statement-II is correct

OFFICIAL LOCAL SET-A KEY: D.

Statement-level explanation: High-mass giant stars have much faster fusion rates and normally shorter lifespans. Statement II is correct; Statement I reverses the mass-lifetime relationship.

37. UPSC GS-I 2024 Q15 - auroras

Exact local-paper wording: “What are aurora australis and aurora borealis? How are these triggered?” (15 marks, 250 words)

Independent model solution

Aurora borealis and aurora australis are luminous upper-atmospheric displays around the northern and southern geomagnetic poles respectively. They are conjugate expressions of solar-terrestrial coupling. The Sun continuously emits charged particles as solar wind; coronal mass ejections can intensify this flow. Earth's magnetosphere deflects most particles, but magnetic reconnection in the magnetotail accelerates some along converging field lines into polar auroral ovals. Collisions with oxygen and nitrogen excite atmospheric atoms and molecules; de-excitation emits light. Oxygen commonly produces green near 100-200 km and red at higher altitude, while nitrogen adds blue, purple and pink at lower levels. The ovals are centred on magnetic, not geographic, poles and can expand equatorward during strong geomagnetic storms. The same process can disturb HF radio, GNSS, satellites, power grids and polar aviation. Thus aurora is not reflected sunlight or tropospheric weather; it is the visible signature of a dynamic magnetic shield interacting with space weather.

Why this earns marks: It defines both terms, supplies a complete causal chain, explains location and colour, adds consequences and explicitly removes common misconceptions.

38. UPSC Prelims 2025 Q33 - rotation/axis, solar flares and polar ice

Exact local-paper wording: Consider the following statements:

Statement I: Scientific studies suggest that a shift is taking place in the Earth's rotation and axis. Statement II: Solar flares and associated coronal mass ejections bombarded the Earth's outermost atmosphere with tremendous amount of energy. Statement III: As the Earth's polar ice melts, the water tends to move towards the equator.

- A. Both Statement II and Statement III are correct and both of them explain Statement I
- B. Both Statement II and Statement III are correct but only one of them explains Statement I
- C. Only one of the Statements II and III is correct and that explains Statement I
- D. Neither Statement II nor Statement III is correct

OFFICIAL LOCAL SET-A KEY: B.

Statement-level explanation: Statements II and III are correct. Redistribution of mass from polar ice toward lower latitudes can influence rotation and polar motion; solar energy deposition in the outer atmosphere does not supply that mass-redistribution explanation.

39. UPSC Prelims 2025 Q77 - Anadyr and Nome

Exact local-paper wording: Consider the following statements:

I. Anadyr in Siberia and Nome in Alaska are a few kilometers from each other, but when people are waking up and getting set for breakfast in these cities, it would be different days. II. When it is Monday in Anadyr, it is Tuesday in Nome.

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

OFFICIAL LOCAL SET-A KEY: A.

Statement-level explanation: Statement I is correct because the places lie on opposite sides of the International Date Line. Statement II is reversed: Anadyr is west of the line and normally a calendar day ahead of Nome, so Monday in Anadyr corresponds broadly to Sunday in Nome at similar local times.

40. Six original solved Mains models

Original 10-marker - Explain why Earth-Sun distance does not cause the seasons.

Model solution

Seasons result primarily from Earth's 23.5° axial tilt acting through revolution, not from changing distance. As Earth orbits the Sun, the hemisphere tilted sunward receives a higher solar angle and longer daylight, raising daily insolation; the opposite hemisphere receives oblique rays and shorter days. This is why the hemispheres experience opposite seasons at the same Earth-Sun distance. Named evidence strengthens the mechanism: the June solstice places the overhead Sun at the Tropic of Cancer and gives long Northern Hemisphere days, whereas the December solstice reverses the pattern. Earth is also near perihelion in early January, during Northern Hemisphere winter, directly contradicting a simple distance explanation. Distance slightly modifies received energy, but it cannot explain opposite hemispheric seasons. Therefore axial geometry, not proximity, is the controlling cause.

Why this earns marks: The answer directly rejects the misconception, explains solar angle and day length, uses solstice and perihelion evidence, and gives a qualified verdict.

Original 10-marker - Explain why India's similar angular spans produce unequal linear dimensions.

Model solution

Angular equality on a globe does not create equal linear distance because latitude and longitude have different geometry. Mainland India spans about 29°02' in latitude and 29°18' in longitude, yet measures roughly 3,214 km north-south and 2,933 km east-west. A degree of latitude remains nearly constant because parallels are measured along meridians. Meridians, however, converge toward the poles, so the linear length of a longitude degree equals approximately 111 km multiplied by the cosine of latitude. India's east-west spread therefore traverses longitude degrees shorter than at the Equator. The country's irregular outline also means quoted dimensions do not equal the sides of a perfect coordinate rectangle. Hence coordinate span must be converted through spherical geometry rather than read as a flat grid.

Why this earns marks: Exact figures serve the mechanism, the cosine/convergence principle is stated, and the irregular-outline qualification prevents false precision.

Original 15-marker - Explain how seismic waves reveal Earth's differentiated interior.

Model solution

Earth's interior cannot be directly observed at depth, so seismic waves act as probes. P-waves are compressional and travel through solids and liquids; S-waves are shear waves and travel only through solids. Their velocities change and paths refract when density and elastic properties change, identifying discontinuities. The Moho separates crust and mantle; the Gutenberg discontinuity at about 2,900 km marks the mantle-outer-core boundary. S-waves disappear beyond this boundary, demonstrating that the outer core is liquid. P-waves slow and refract sharply, producing a P-wave shadow zone rather than vanishing completely. At greater depth, changes in P-wave behaviour across the Lehmann boundary support a solid inner core under enormous pressure. Surface waves explain much earthquake damage but do not map the deep interior in the same way. Thus the combined transmission, non-transmission, reflection, refraction and shadow-zone pattern reveals a layered, compositionally and mechanically differentiated planet, though precise values improve as global seismic networks and models advance.

Why this earns marks: It uses named waves, named discontinuities and shadow zones as evidence, links each to an inference, and adds a methodological qualification.

Original 15-marker - Analyse how India's location has shaped climate, cultural contacts and maritime strategy.

Model solution

India's location supplies an enabling physical and relational frame, but outcomes arise through circulation, routes and state capacity. Tropical-subtropical latitude, the Himalayan arc and Arabian Sea-Bay of Bengal moisture establish the setting for seasonal monsoon circulation and climatic contrasts. The Himalaya filters cold continental air and forces uplift, while the ocean moderates coasts; however relief and circulation, not latitude alone, explain rainfall. North-western passes, Himalayan valleys and long coasts linked India with Central Asia, West Asia, East Africa and South-East Asia, enabling multidirectional movement of crops, faiths, technologies and communities. The peninsula and island chains also place India near northern Indian Ocean routes connecting Gulf/Red Sea approaches with South-East Asia, supporting ports, energy security, maritime awareness and disaster response. Yet India does not automatically control Hormuz, Bab-el-Mandeb or Malacca, and strategic use must respect island ecology and communities. Location is therefore potential converted into outcomes by infrastructure, institutions and cooperation.

Why this earns marks: Three dimensions are integrated through named spatial evidence, causal mechanisms and an anti-determinist strategic qualification.

Original 20-marker - Trace the planetary sequence from the Big Bang to a habitable Earth and assess its geographic significance.

Model solution

Planetary geography begins at cosmic scale but becomes meaningful only when each stage explains an Earth-system property. The Big Bang model describes expansion and cooling from a hot, dense early universe about 13.8 billion years ago; redshift, the cosmic microwave background and light-element abundance provide mutually reinforcing evidence. Gravity concentrated matter into galaxies and stars. Inside stars, fusion created energy and, over stellar generations, many heavier elements required for rocky planets. Mass controlled stellar lifespans and endpoints, while supernovae dispersed enriched material. About 4.6 billion years ago a rotating gas-dust cloud collapsed to form the Sun and a protoplanetary disc. Temperature gradients and accretion differentiated rocky inner planets from giant outer planets. Earth then developed a differentiated crust-mantle-core structure, a gravity field expressed by the geoid, rotation and revolution, an atmosphere-hydrosphere system and a magnetosphere generated by core dynamics. Its orbital position permits liquid water, but habitability is not distance alone: atmosphere, magnetic shielding, geochemical cycling and long-term stability matter. The sequence is therefore not a deterministic ladder toward life; it explains the material and physical preconditions that make Earth's geography distinctive.

Why this earns marks: The answer links four scales, uses three Big Bang evidence units, explains element and planet formation, names Earth-system outcomes and avoids teleology.

Original 20-marker - "The Himalaya and the Indian Ocean have made India both a relatively coherent geographic unit and an open arena of exchange." Discuss.

Model solution

The apparent contradiction captures India's locational logic: mountain-ocean enclosure creates relative coherence, while selective gateways sustain exchange. The Himalayan-Karakoram arc separates the subcontinent from the Tibetan and Central Asian interiors, checks very cold continental air and helps organise river and monsoon systems. This gives the northern plains and peninsula a shared environmental frame. Yet passes and valleys were never sealed; they carried trade, pilgrimage, pastoral mobility and political contact. To the south, the peninsula is flanked by the Arabian Sea and Bay of Bengal and extended by Lakshadweep and Andaman-Nicobar. Maritime movement connected western India to Gulf-East African circuits and eastern India to Sri Lanka and South-East Asia; historically, monsoon winds structured sailing seasons. Modern shipping links West Asian energy routes with East Asian production networks through the wider Indian Ocean. However, external chokepoints remain beyond Indian sovereignty, while coasts and islands face cyclones, tsunamis, erosion, ecological limits and community concerns. Mountains and seas are therefore filters and connectors, not deterministic walls or highways. India's geographic unity is relative and relational: enclosure supplies a recognisable subcontinental frame, while permeability explains cultural plurality and maritime significance.

Why this earns marks: The answer resolves the quotation, balances unity with permeability, uses named spatial evidence across historical and modern scales, and provides a graded verdict.